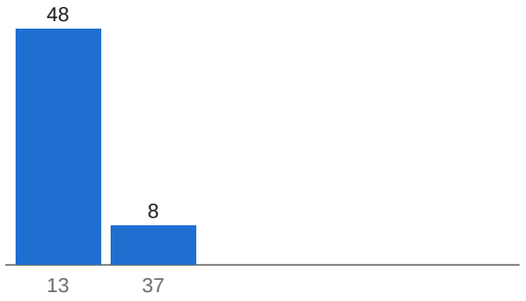
GHS cover census, n = 4 (K = F_(q^4)): minimal admissible genus per curve, by 2-torsion class

measured counts of curve isomorphism classes; x axis: the curve's minimum genus over all F_q-structures (sigma-orbit length 4); run 20261024

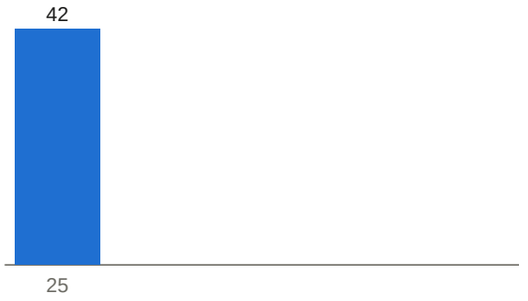
P1 q = 3, exhaustive (all curve classes)

#E odd



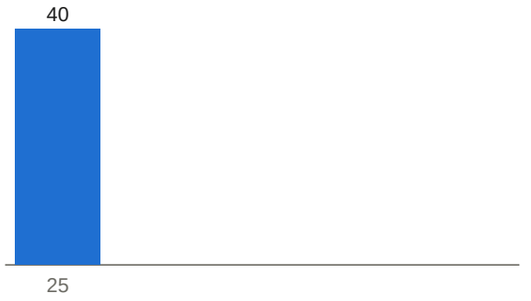
n = 56 curves; class min g = 13
model exponent 2 - 2/g = 1.846 (rho: 2)

#E = 2 mod 4 (E[2](K) = Z/2)



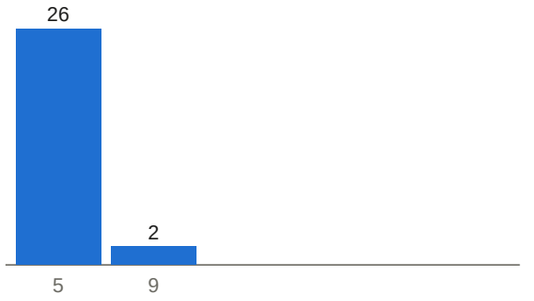
n = 42 curves; class min g = 25
model exponent 2 - 2/g = 1.920 (rho: 2)

4 | #E, E[2](K) = Z/2



n = 40 curves; class min g = 25
model exponent 2 - 2/g = 1.920 (rho: 2)

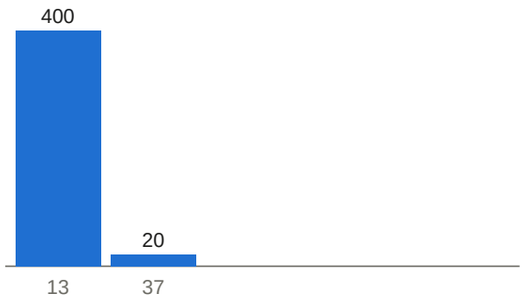
E[2](K) = (Z/2)^2



n = 28 curves; class min g = 5
model exponent 2 - 2/g = 1.600 (rho: 2)

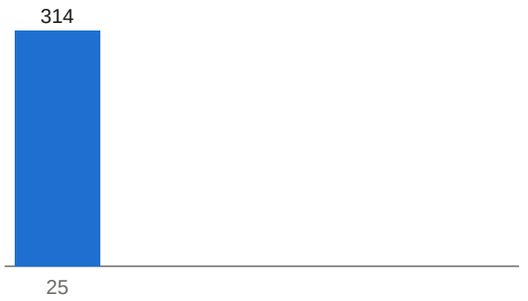
P2 q = 5, exhaustive (all curve classes)

#E odd



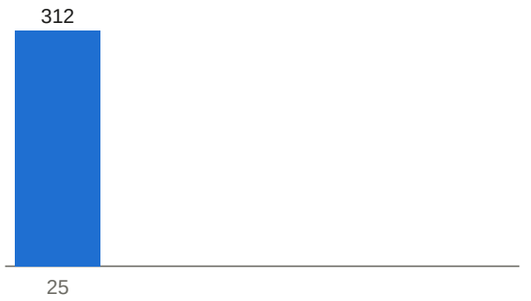
n = 420 curves; class min g = 13
model exponent 2 - 2/g = 1.846 (rho: 2)

#E = 2 mod 4 (E[2](K) = Z/2)



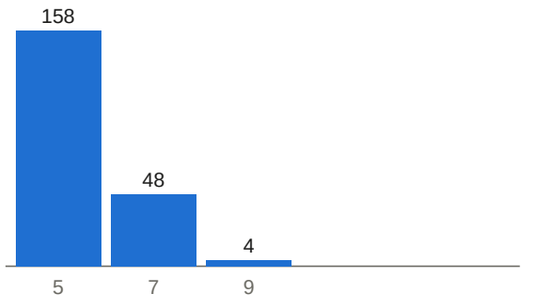
n = 314 curves; class min g = 25
model exponent 2 - 2/g = 1.920 (rho: 2)

4 | #E, E[2](K) = Z/2



n = 312 curves; class min g = 25
model exponent 2 - 2/g = 1.920 (rho: 2)

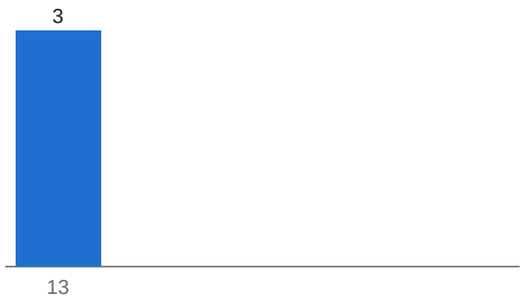
E[2](K) = (Z/2)^2



n = 210 curves; class min g = 5
model exponent 2 - 2/g = 1.600 (rho: 2)

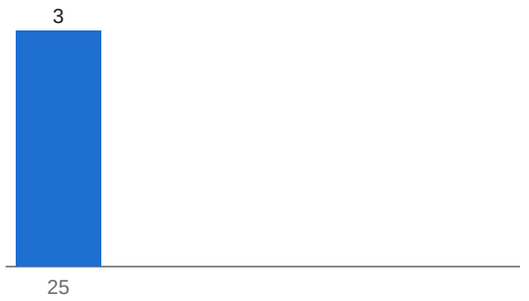
P3 q = 7, seeded sample of 12 curves (neighbour models not in these counts)

#E odd



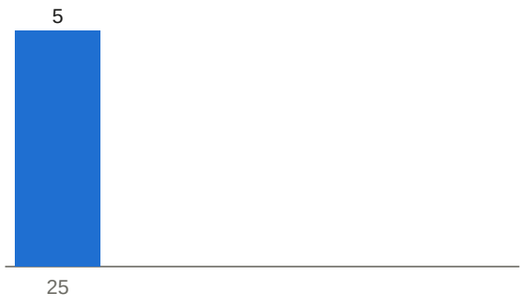
n = 3 curves; class min g = 13
model exponent 2 - 2/g = 1.846 (rho: 2)

#E = 2 mod 4 (E[2](K) = Z/2)



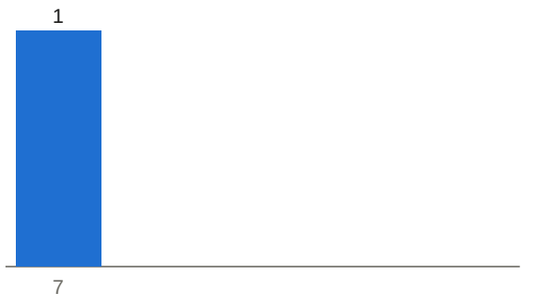
n = 3 curves; class min g = 25
model exponent 2 - 2/g = 1.920 (rho: 2)

4 | #E, E[2](K) = Z/2



n = 5 curves; class min g = 25
model exponent 2 - 2/g = 1.920 (rho: 2)

E[2](K) = (Z/2)^2



n = 1 curves; class min g = 7
model exponent 2 - 2/g = 1.714 (rho: 2)

Model exponents are asymptotic formulas (constants ~g! and descent costs not included), not measurements. Source: results/run-20261024/*.jsonl